

Alarm Summary Page

Ignition Perspective build instruction for the two-tank batch-mixing station

A first-out banner, a twelve-row alarm table built from one reusable row view, live/sealed/alert indicators per alarm, Reset and Silence, page navigation, and the Gateway heartbeat that keeps alarm 11 clear.

Design decision

The PLC owns every alarm. SBR-ALM detects, delays, seals, and silences; the page only **reads** that state and never computes an alarm of its own. The one exception is Reset and Silence, which write the same request bits the Overview buttons already write. The twelve rows are **not** twelve hand-built groups of components: a single view, **AlarmRow**, is drawn once and repeated twelve times by a Flex Repeater, each copy receiving one alarm's tag-name key as a parameter. Because the PLC's alarm block is index-aligned – condition, sealed alarm, and alert for alarm i always share one name – one parameter is enough for a row to find all three of its tags. Adding a thirteenth alarm is one new entry in a list, not a new row of components.

1 Requirements and Tags

The Overview screen already shows that the station is *Faulted*, but not which alarm caused it, whether the cause is still present, or whether Reset will succeed. The Alarm page answers all three:

Which alarm tripped first – a banner driven by the first-out register DS60, hidden when no code is held.

Which alarms are sealed – one row per alarm with three indicators: **LIVE** the condition is present now, **SEALED** the alarm is locked in, and **ALERT** the alarm is still sounding the horn. An inactive indicator shows **grey**.

Whether Reset will hold – a sealed alarm whose LIVE indicator is grey clears on Reset; one whose LIVE indicator is red re-seals after its delay.

Reset and Silence on the page, plus navigation between Overview and Alarms on every page.

1.1 Tags used by the page

The tags sit at the root of the **default** provider with flat, underscore-separated names – **Alarms_Sealed_CommLoss**, not **Alarms/Sealed/CommLoss**. Every path in this document uses that form. **<Name>** stands for the twelve name keys listed in Section 1.2.

Tag	CLICK	Modbus	R/W	Used for
Alarms_FirstOutCode	DS60	400060	R	First-out banner
Alarms_AnyAlarm	C57	16441	R	Alarm indicator in the header
Alarms_Conditions_<Name>	C600--C611	16984--16995	R	LIVE indicator
Alarms_Sealed_<Name>	C640--C651	17024--17035	R	SEALED indicator
Alarms_Alerts_<Name>	C660--C671	17044--17055	R	ALERT indicator
Commands_Reset	C25	16409	W	Reset button
Commands_Silence	C26	16410	W	Silence button
Comms_HeartbeatCounter	DS51	400051	W	Gateway heartbeat (Section 5)

The three command/heartbeat names follow the same flat convention; confirm the exact spelling in the Tag Browser before typing them into a script. Right-click any tag → **Copy Path** and paste it into a text editor to see the exact form Ignition expects.

1.2 Alarm list

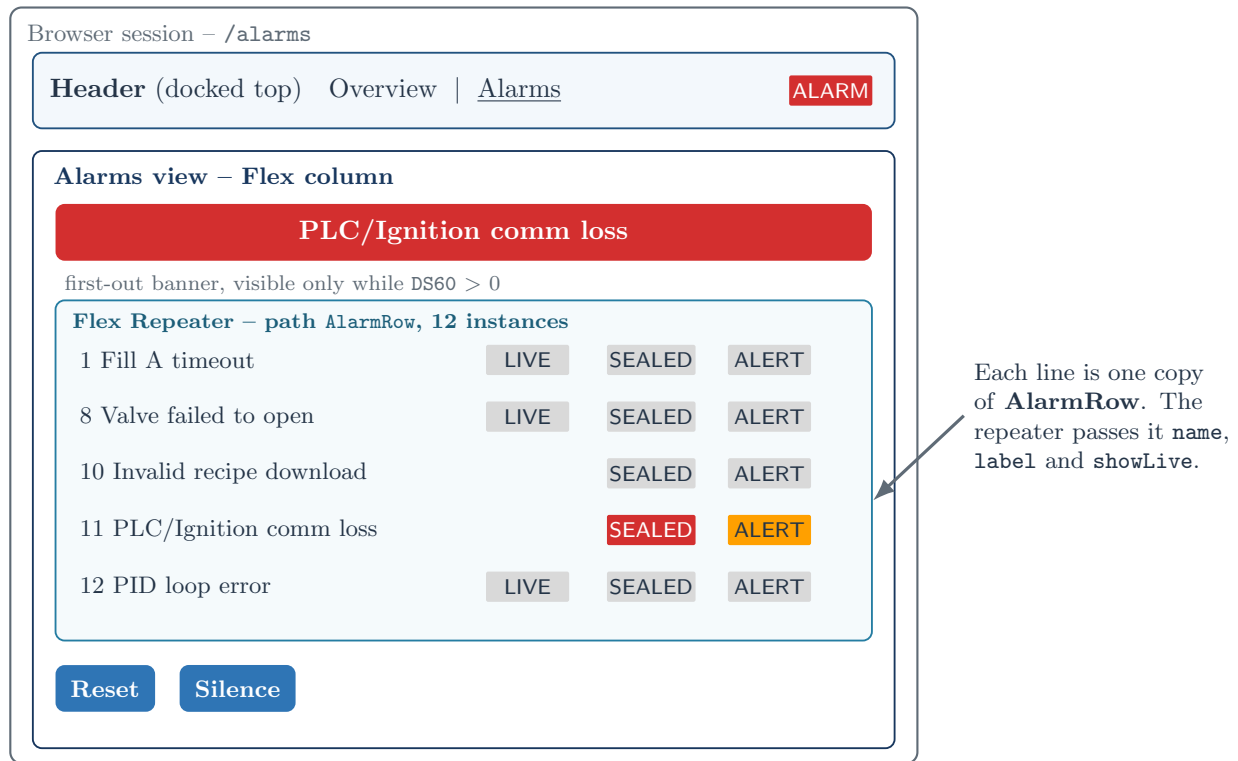
#	Name key	Row label	Cond	Sealed	Alert	showLive
1	FillATimeout	Fill A timeout	C600	C640	C660	true
2	FillBTimeout	Fill B timeout	C601	C641	C661	true
3	TransferTimeout	Transfer timeout	C602	C642	C662	true
4	HeatTimeout	Heat step timeout	C603	C643	C663	true
5	DischargeTimeout	Discharge timeout	C604	C644	C664	true
6	HeaterOvertemp	Heater overtemperature	C605	C645	C665	true
7	MixTankHighHigh	Mix tank high-high	C606	C646	C666	true
8	ValveFeedbackFault	Valve failed to open	C607	C647	C667	true
9	PumpFeedbackFault	Pump without feedback	C608	C648	C668	true
10	InvalidRecipe	Invalid recipe download	C609	C649	C669	false
11	CommLoss	PLC/Ignition comm loss	C610	C650	C670	false
12	PidLoopError	PID loop error	C611	C651	C671	true

Why LIVE is hidden on rows 10 and 11, and what LIVE means on rows 1–5

Two condition bits cannot be displayed honestly at Ignition's poll rate. C609 is true for exactly one scan – the scan a download is rejected – so a poll almost never catches it. C610 is /C111, which is true on every scan except the one where the heartbeat advances, so it would read red nearly all the time with communications perfectly healthy. Both rows therefore hide their LIVE indicator through the **showLive** parameter; their SEALED and ALERT indicators are fully valid.

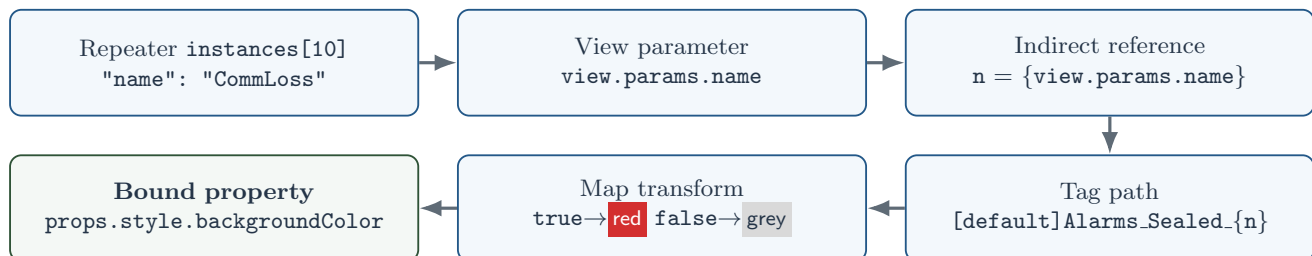
On rows 1–5 the condition is simply “this step is active.” Once the sequencer faults, DS10 is 999 and those conditions are false, so LIVE reads grey after every timeout trip. That is correct and useful: it tells the operator the timeout will clear on Reset.

2 Page Structure



View	Page URL	Purpose
Overview	/	The existing process screen. Opens first when a session starts.
Alarms	/alarms	Alarm summary. First-out banner, the Flex Repeater of twelve rows, Reset and Silence.
AlarmRow	none	One alarm row. A description label and three indicators, each found through the name parameter. Only ever used inside the repeater, so it has no page URL.
Header	docked	Navigation. Horizontal Menu and the any-alarm indicator, docked to the top of every page through the shared page settings.

2.1 How one indicator finds its tag



Every indicator follows this chain; only the folder word in the tag path differs (**Conditions**, **Sealed**, or **Alerts**). Each link is a place a binding can fail, and Section 3.5 lists the error each broken link produces.

3 Building the Views

The steps below are in build order. **AlarmRow** comes first because the repeater in **Alarms** needs it to exist. Save the project (**Ctrl+S**) at the end of each subsection.

3.1 Create the AlarmRow view

Step 1 – New view, flex row

1. In the **Project Browser**, expand **Perspective**, right-click **Views** → **New View...**
2. **View Name** **AlarmRow**, **Root Container Type** **Flex Container**. Leave **Page URL** unchecked. Click **Create View**.
3. Select **root**. In **PROPS**, set **direction** to **row** and **alignItems** to **center**.
4. Select the view node **AlarmRow** (the node above **root**). In **PROPS** → **defaultSize**, set **height** to 40 and **width** to 600.

Step 2 – Three view parameters

With the view node still selected, use **PARAMS** → **Add View Parameter...** three times, choosing **Value** each time. **Object** and **Array** are wrong here; an entry that shows **[0]** or **{...}** beside its name was created as the wrong type and should be deleted.

Name	Default value	Purpose
name	ValveFeedbackFault	Tag-name key used to build all three tag paths
label	8 Valve failed to open	Text shown at the left of the row
showLive	true	Hides the LIVE indicator when false

The defaults exist only so the row previews in the Designer; the repeater overrides all three per row. Leave each parameter's direction arrow at **Input**.

3.2 The description label

Step 3 – Label bound to the label parameter

1. Drag a **Label** from the Perspective Components palette into the view.
2. **POSITION**: **grow** 0, **shrink** 0, **basis** 250px.
3. Click the link icon beside **props.text** → **Property** → **view.params.label** → **OK**. The label should read *8 Valve failed to open*.

A **Property** binding takes the path without braces. Braces are needed only where a field is evaluated as an expression (Step 4).

3.3 The LIVE indicator

Step 4 – Label named Condition

1. Drag in a second **Label**. Set `meta.name` to `Condition` and `props.text` to `LIVE`.
2. **POSITION**: grow 0, shrink 0, basis 80px.
3. Set `props.textStyle.textAlign` to `center`. Under `props.style` add `margin 2px` and `borderRadius 4px`.
4. Expand `props` → `style`, click + **Add Property...**, and name it `backgroundColor`. It must sit *indented under style*, below `classes`, and the count shown beside `style` goes up by one.
5. Click the link icon beside that `style.backgroundColor` and configure a **Tag** binding:

Select **Indirect**.

Tag Path: `[default]Alarms_Conditions_{n}`

In the reference table add `n`. Double-click its **Property** cell, type `{view.params.name}` and press **Enter**, or use the cell's **fx** button to pick `view` → `params` → `name`.

6. **Add Transform** + → **Map**. Set **Input Type** to **Value** and **Output Type** to **Color**, then fill:

Input	Output
true	#D32F2F red
false	#D9D9D9 light grey
Fallback	#9E9E9E grey

7. Click **Apply**. The dialog title must read `Condition.props.style.backgroundColor`, and **Binding Preview** must show `false` → `#D9D9D9` with no red icon. Click **OK**; the badge turns light grey on the canvas.
8. Click the link icon beside `meta.visible` → **Property** → `view.params.showLive` → **OK**.

3.4 The SEALED and ALERT indicators

Step 5 – Two copies of Condition

Select **Condition**, **Ctrl+C**, select **root**, **Ctrl+V**, and repeat once more. Change each copy as follows:

Copy	meta.name	props.text	Tag Path	Map true
First	Sealed	SEALED	<code>[default]Alarms_Sealed_{n}</code>	#D32F2F
Second	Alert	ALERT	<code>[default]Alarms_Alerts_{n}</code>	#FFA000

On both copies, open `meta.visible`, click **Remove Binding**, and confirm `visible` shows `true`. **SEALED** and **ALERT** are meaningful for all twelve alarms and must never hide.

After pasting, open each copy's `style.backgroundColor` binding and check two things before changing the path: that the binding is still under `style`, and that the `n` cell still holds `{view.params.name}`. A paste can drop either.

3.5 Binding faults and what they mean

Symptom	Cause and fix
Preview is good, but the badge shows no colour at all, not even grey	The binding is on <code>props.backgroundColor</code> , which a Label ignores. Right-click it → Copy Binding , add <code>backgroundColor</code> under style , Paste Binding , then delete the stray property. The dialog title is the quickest check.
<code>Error_Configuration</code> with no further detail	The <code>n</code> reference cell is empty. The cell only keeps its value after Enter ; typing and then clicking Apply or OK discards the edit.
<code>Error_Configuration: Illegal character <.> (Line 1, Char 5)</code>	The reference cell holds <code>view.params.name</code> without braces. That cell is an expression; enter <code>{view.params.name}</code> .
<code>Bad_NotFound: Path ... not found</code>	The path builds but names no tag. The usual causes are slashes typed instead of underscores, a misspelled name key, or a different provider name. Compare against Copy Path .
<code>Bad_Stale</code> or bad quality on every tag	The device connection is down. Check Config → OPC UA → Device Connections on the Gateway web page.
Every badge shows the grey fallback	The path resolves to nothing or the value is not boolean. Confirm the tag's data type is Boolean .

Test the finished row in the Designer

Select the **AlarmRow** view node and set **name** to the key of an alarm that is currently sealed – for example **CommLoss** while the heartbeat is not running. **SEALED** should turn red and **ALERT** amber on the canvas immediately. Set **name** back to **ValveFeedbackFault** afterwards; the value is only a preview default.

3.6 Create the Alarms view

Step 6 – New view with a page URL

1. **Views** → **New View...**, name **Alarms**, root **Flex Container**.
2. Tick **Page URL** and enter `/alarms`. Click **Create View**.
3. Select **root** and confirm **direction** is **column**.
4. Select the view node and set **defaultSize** → **height** to 600.

Step 7 – First-out banner

Drag in a **Label**, set **meta.name** to `FirstOutBanner`, and style it: `style.backgroundColor #D32F2F`, `style.color #FFFFFF`, `style.padding 6px`, `textStyle.fontWeight bold`. Bind **props.text** with an **Expression** binding:

```
case([[default]Alarms_FirstOutCode],
  1,"Fill A timeout", 2,"Fill B timeout", 3,"Transfer timeout",
  4,"Heat step timeout", 5,"Discharge timeout", 6,"Heater overtemperature",
  7,"Mix tank high-high", 8,"Valve failed to open", 9,"Pump without feedback",
  10,"Invalid recipe download", 11,"PLC/Ignition comm loss", 12,"PID loop error",
  "")
```

Bind **meta.visible** with a second **Expression** binding:

```
[[default]Alarms_FirstOutCode] > 0
```

The final "" is the default for any code outside 1–12. The quickest way to insert the tag reference without a typo is to select it and use the tag-browse button beside the editor. To test the expression apart from the tag, temporarily replace the reference with the literal 8; the preview should read *Valve failed to open*.

3.7 The Flex Repeater

Step 8 – Repeater properties

Drag a **Flex Repeater** into **Alarms** below the banner and set:

Property	Value	Why
<code>props.path</code>	<code>AlarmRow</code>	The view drawn for each instance
<code>props.direction</code>	<code>column</code>	Rows stack vertically
<code>props.elementPosition.grow</code>	0	Rows keep their own height ...
<code>props.elementPosition.shrink</code>	0	... and are never squeezed
<code>props.elementPosition.basis</code>	40px	Matches the <code>AlarmRow</code> height
<code>position.grow</code>	1	The repeater fills the space under the banner
<code>position.basis</code>	<code>auto</code>	
<code>props.style.overflow</code>	<code>auto</code>	A short window scrolls instead of cutting rows off

Twelve 40 px rows need 480 px. If the repeater is ever placed inside another container, that container needs `position.grow 1` as well; the smallest container between **root** and the repeater sets the visible height.

Step 9 – Instances from a script transform

`props.instances` starts as an empty array. The most reliable way to fill it is a binding:

1. Click the link icon beside `instances` → **Expression**, and enter the placeholder 1.
2. **Add Transform** + → **Script**. Leave the generated `def transform(...):` line untouched and replace `return value` with the block below, indented one level exactly as `return value` was.
3. The preview should show an array of 12. Click **OK**.

```
return [
    {"name": "FillATimeout", "label": "1 Fill A timeout", "showLive": True},
    {"name": "FillBTimeout", "label": "2 Fill B timeout", "showLive": True},
    {"name": "TransferTimeout", "label": "3 Transfer timeout", "showLive": True},
    {"name": "HeatTimeout", "label": "4 Heat step timeout", "showLive": True},
    {"name": "DischargeTimeout", "label": "5 Discharge timeout", "showLive": True},
    {"name": "HeaterOvertemp", "label": "6 Heater overtemperature", "showLive": True},
    {"name": "MixTankHighHigh", "label": "7 Mix tank high-high", "showLive": True},
    {"name": "ValveFeedbackFault", "label": "8 Valve failed to open", "showLive": True},
    {"name": "PumpFeedbackFault", "label": "9 Pump without feedback", "showLive": True},
    {"name": "InvalidRecipe", "label": "10 Invalid recipe download", "showLive": False},
    {"name": "CommLoss", "label": "11 PLC/Ignition comm loss", "showLive": False},
    {"name": "PidLoopError", "label": "12 PID loop error", "showLive": True}
]
```

The list is Python, so `True` and `False` are capitalised. A red preview after pasting is almost always indentation: every line must use the same indent character as the line it replaced. To add or rename an alarm later, reopen this binding and edit the list.

3.8 Reset and Silence

Step 10 – Reuse the Overview buttons

Copy the **Reset** and **Silence** buttons from the Overview view and paste them into Alarms below the repeater, inside a small Flex row container. Copying keeps the event script already proven against the PLC's request handling, so both screens release `Commands_Reset` and `Commands_Silence` the same way. Do not write a second, different script for the same two bits.

4 Navigation Between Pages

Step 11 – Page Configuration

Open **Perspective** → **Page Configuration** (double-click it in the Project Browser).

1. Confirm `/` maps to the Overview view; whatever is on `/` opens first in a new session.
2. Confirm `/alarms` maps to **Alarms**. If it is missing, click `+`, enter the URL, and choose the view as **Primary View**.
3. To rename a URL later, select it here and edit the **URL** field. Update the matching menu target in Step 12 at the same time.

Step 12 – Header view with a menu and an alarm indicator

1. Create a view **Header**: Flex root, `direction row`, `alignItems center`, `defaultSize` height 50. No page URL.
2. Drag in a **Horizontal Menu** with `position.grow 1`. Replace its `props.items` so it holds two entries:

```
[
  {"label": "Overview", "target": "/"},
  {"label": "Alarms", "target": "/alarms"}
]
```

If pasting JSON is not offered, add the two objects with **Add Array Element...** → **Object** and give each a `label` and `target` value.

3. Drag in a **Label** with text **ALARM**, `basis 80px`, centred. Bind `style.backgroundColor` to the **Tag** `[default]Alarms_AnyAlarm (Direct)` with a Map transform: `true` → `#D32F2F`, `false` → `#D9D9D9`. The operator now sees a fault from any page.
4. In **Page Configuration** → **Shared Settings** → **Docked Views**, add **Header** with anchor **top** and display **visible**.

Buttons with an **onClick** → **Navigation** action set to a Page URL work equally well in place of the menu. Recipe, Trends, and Diagnostics pages are added the same way later: a new view with a URL, and one more menu entry.

5 Keeping Alarm 11 Clear: the Gateway Heartbeat

Alarm 11 seals whenever the PLC sees no change in DS51 for 5 s. DS51 is Ignition-writable, but nothing writes it until a script is added – so without the step below, alarm 11 re-seals five seconds after every power-up and every Reset, and the Alarm page will always show it.

Step 13 – Gateway timer script

1. Project Browser → **Scripting** → **Gateway Events** → **Timer** tab → +.
2. **Name** Heartbeat, **Delay** 1000 ms, **Delay Type** Fixed Rate, **Enabled** ticked.
3. Paste the script, adjusting the path to the exact name shown by **Copy Path**:

```
path = "[default]Comms_HeartbeatCounter"
v = system.tag.readBlocking([path])[0].value
if v is None:
    v = 0
system.tag.writeBlocking([path], [(v + 1) % 30000])
```

The script runs on the Gateway, so the heartbeat continues with no client session open. The modulo keeps the value inside the Int2 range; the wrap from 29 999 to 0 is still a change, so the PLC never sees a stall at rollover.

If alarm 11 still re-seals with the counter moving

Watch **Comms_HeartbeatCounter** in the Tag Browser: it should step once per second. If it does, open CLICK Data View and confirm DS51 shows the same changing value. A counter that moves in Ignition but not in the PLC means the tag is not really writing – check that it is not marked read-only in the tag provider and that its OPC item path is **[CLICK]HR51**.

6 How the Page Reads

Each indicator is one PLC bit. Read together, a row states both what happened and what Reset will do:

LIVE	SEALED	ALERT	Meaning
LIVE	SEALED	ALERT	Healthy. Nothing to do.
LIVE	SEALED	ALERT	Condition present, still inside its delay. The alarm will seal if it persists.
LIVE	SEALED	ALERT	Tripped, cause has gone. Silence quiets the horn; Reset will clear it.
LIVE	SEALED	ALERT	Tripped and silenced. The sequencer is still faulted; Reset will clear it.
LIVE	SEALED	ALERT	Tripped and the cause is still present. Reset drops it for one scan and it re-seals after its delay – fix the cause first.

First out: the banner names the alarm that sealed first, even when several are sealed. It clears only once Reset has succeeded in dropping every sealed alarm.

Silence: ALERT goes grey and the horn stops; SEALED stays red and the station stays faulted. A quiet panel is never a cleared one.

Reset: SEALED and ALERT clear on every row whose condition is gone. Any row whose condition is still live returns after its delay.

Rows 10 and 11 show no LIVE badge by design (Section 1.2); read SEALED and ALERT only.

7 Verification Against the Test Criteria

Test action	Expected result
Open a session and select <i>Alarms</i> in the header	The page opens at <code>/alarms</code> ; all twelve rows are visible; rows 10 and 11 show no LIVE badge.
Heartbeat running, press Reset	Every badge grey, banner hidden, header ALARM grey.
Disable the Heartbeat timer script	Within 5 s row 11 shows SEALED red and ALERT amber; banner reads <i>PLC/Ignition comm loss</i> ; header ALARM turns red.
Press Silence	Row 11 ALERT turns grey and the horn stops; SEALED stays red; banner stays.
Press Reset with the timer still disabled	Row 11 clears for one scan and re-seals 5 s later. It must not stay clear.
Re-enable the timer, press Reset	Row 11 and the banner clear and stay clear.
Throw X008 during Fill A	Row 8 LIVE turns red at once; after 2 s SEALED red and ALERT amber; banner reads <i>Valve failed to open</i> ; LIVE turns grey as DS10 goes to 999.
Press Reset with X008 still thrown	Row 8 clears and stays clear, because the valve command dropped with the fault. The next Start re-trips it after 2 s.
Force T1 to expire during Fill A	Row 1 SEALED and ALERT; LIVE grey once faulted; banner reads <i>Fill A timeout</i> .
Designer: set <code>AlarmRow.name</code> to a sealed alarm's key	The row's badges colour on the canvas without launching a session.
Shrink the browser window below 500 px high	The repeater scrolls and all twelve rows remain reachable.
Select <i>Overview</i> in the header	The Overview page opens with the header still docked on top.

8 Implementation Checks

Before relying on the page, confirm the following:

1. Every indicator binding's dialog title ends in `.props.style.backgroundColor`, and no `backgroundColor` remains directly under `props` on any label.
2. Every indirect reference cell reads `{view.params.name}`, with braces, on all three indicators – reopen each binding to confirm it survived.
3. Every tag path uses underscores and matches **Copy Path** output exactly: `Alarms.Conditions_{n}`, `Alarms.Sealed_{n}`, `Alarms.Alerts_{n}`.
4. Every Map transform has Output Type **Color** and a Fallback row, so a bad tag shows grey rather than an error.
5. `AlarmRow` has exactly three **Value** parameters, all **Input**, and no page URL.
6. Only the **Condition** label's `meta.visible` is bound; **Sealed** and **Alert** are always visible.
7. The repeater's `instances` binding previews an array of exactly 12, and rows 10 and 11 carry `showLive: False`.
8. The page writes only `Commands.Reset` and `Commands.Silence`, through buttons copied from Overview. No binding on the page is bidirectional.

9. **Page Configuration** maps / and /alarms, and **Header** is docked top in Shared Settings.
10. The **Heartbeat** Gateway timer is enabled, and **Comms_HeartbeatCounter** steps once per second in both Ignition and CLICK Data View.
11. No alarm logic exists in Ignition. Ignition tag alarms, if added later for history and timestamps, are configured on the **Alarms_Sealed_*** tags only, and Ignition acknowledgement never writes **Commands_Reset**.

Result

One reusable row view and one list of twelve entries give the operator every alarm's live, sealed, and alert state, the first-out cause, and a clear answer to "will Reset work?" – while the PLC remains the only place alarms are detected, sealed, and cleared.